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Whether we are discussing individual plasmonic nanostructures, or metamaterials and photonic crystals composed of arrays of dielectric or metallic building blocks, structural designs play a central role.

Such inverse design algorithms enable one to find non-intuitive, irregularly shaped photonic structures that outperform empirically designed structures in many applications, such as silicon photonic components, photonic crystals and metamaterials.

Readers interested in inverse design in photonics can refer to recent reviews on this topic^{9–11}.

In light of its exceptional success in domains related to computer science and engineering, including computer vision¹³, natural language processing¹⁴, speech recognition¹⁵, knowledge graphs¹⁶ and decision making¹⁷, deep learning has attracted increasing attention from researchers in other disciplines, including materials science¹⁸, chemistry¹⁹, laser physics²⁰, particle physics²¹, quantum mechanics²², computational imaging²³ and microscopy²⁴, demonstrating potential to circumvent the drawbacks of traditional methods and create unprecedented opportunities in these areas.

The unique advantages of deep learning lie in its data-driven methodology, which allows the model to discover useful information automatically from a huge amount of data, in sharp contrast to physics- or rule-based approaches.

Over the past few years, deep learning has become a radically new approach in the context of photonic design.

In this Review, we draw attention to a collection of recent results that showcase the power of deep learning in the design of photonic structures, materials and devices, where empirical or traditional approaches are infeasible or inefficient.

We begin by providing background on the deep-learning model for photonics and highlighting the formulation, development and advantages of deep neural networks.

Then we discuss several major model architectures, from the basic multilayer perceptron (MLP) and advanced deep neural networks to hybrid models with other optimization methods, emphasizing their potential to design photonic crystals, metamaterials, plasmonic nanostructures and integrated silicon photonic devices (Fig. 1).

Background

Historically, deep learning can be traced back to the 1940s, and it went through many different names before becoming popularly known by this term²⁵.

Therefore, deep learning was known by the term artificial neural networks (ANNs) since the 1980s, accompanied by the second wave of AI research, which largely emerged through a movement called connectionism²⁶.

The modern era of neural network research began with a breakthrough in 2006: Hinton, who later coined the term deep learning, showed that deep neural networks could be efficiently trained using a strategy called greedy layer-wise pretraining²⁸.

Deep learning then gained broad popularity, as researchers were able to train deeper neural networks than had been possible before, and the importance of depth of the model architecture was theoretically realized. Aided by the ever-increasing scale of available data, deep learning still leads the current boom in AI research, with the performance surpassing previous models by a large margin, or even beating humans²⁹.

In the context of photonics research, the introduction of deep learning (or more appropriately ANNs, at that time) dated back to the 1990s during the second prevalence period of AI research.

The use of ANNs in microwave and other photonic designs follows a standard supervised learning paradigm, which aims to find a mapping between input variable X and output

variable Y . The training process is to optimize the model parameters to make it an accurate representation of the desired mapping, on a previously collected training dataset containing examples of X and its corresponding label Y . Starting with random initialization, this data-driven learning approach modifies the model parameters iteratively on the training dataset according to a specific loss function evaluated on true labels Y , until a convergence occurs and the generalized model is able to predict unseen data.

On the other hand, more complex photonic designs such as plasmonic nanostructures, metamaterials, photonic crystals or silicon photonic devices were still in their infancy, with limited understanding in the community.

These structures and devices, featuring deep sub-wavelength dimensions, large design flexibility, extreme dispersions or complicated performance characteristics, were not tractable to the shallow ANN model at that time.

Related research then stagnated until the recent epoch-making development of deep learning.

The past decade has witnessed the rise of deep learning with unprecedented impact on a plethora of research topics.

With the invention of new training and regularization techniques such as ReLU (rectified linear units) activation³⁶, dropout³⁷ and batch normalization³⁸, it is now feasible to design and train deeper and deeper neural networks that can exploit larger datasets with better performance.

The photonics community also benefited from the rapid advances in deep-learning techniques.

Unlike rule-guided optimization that explores the design space by following certain strategies case by case, deep learning, as a data-driven method, aims to describe the design space holistically, using the training data as samples.

Therefore, with generalization ability within a given design space, deep learning can produce fast and accurate designs without the need for case-by-case, time-consuming numerical calculations.

Well-trained deep-learning models can directly set up a mapping from design to optical properties of target photonic devices, and vice versa.

In addition, deep learning can interact with traditional optimization methods to improve the algorithm performance.

specific and representative examples

In modern deep-learning models with sophisticated and task-specific architectures, MLP often serves as a bottleneck layer to extract meaningful features as a compact representation of high-dimensional data such as images.

Even before the astounding success of deep learning in computer vision¹³, researchers explored MLP in photonics-related domains following the design schemes devised by the microwave community.

In 2018, Itzik Malkiel and co-workers reported a bidirectional MLP-based deep-learning model that could be used to design plasmonic nanostructures⁴³.

The scale of the dataset, input complexity and the depth of the model show that, from a data-driven view, deeper models armed with larger datasets are a potential alternative approach to solve photonic design tasks.

Wei Ma and co-workers reported a deep-learning model with two bidirectional neural networks that were integrated by an ensembled learning strategy to achieve on-demand design of chiral metamaterials, which exhibit distinct responses when incident light is left-circularly polarized (LCP) or right-circularly polarized (RCP)⁴⁴.

They also made a quantitative comparison between a deep neural network and interior-point methods, a numerical inverse design algorithm suitable for nanoparticles.

These results indicate that within the same accuracy standard, neural networks, once fully trained, beat conventional numerical inverse design methods by a large margin in speed. Following these first forays into applying data-driven methodology in photonics research, deep MLP models and their variants have been extensively studied for many photonic design tasks, including topological photonics⁴⁷, integrated silicon photonics⁴⁸, colour generation from nanostructures^{49,50}, metamaterials⁵¹⁻⁵³, plasmonic structures⁵⁴ and photonic crystals⁵⁵.

In work shown in Fig. 2e, the researchers propose to use deep learning to design silicon-on-insulator-based 1×2 power splitters with various target transmission ratios at two ports (T_1 and T_2), in which the device structure is represented by a matrix of binary variables, indicating whether the silicon is etched or not at certain coordinates⁴⁸.

In addition to design problems, researchers have also developed sophisticated MLP models to tackle other problems in photonics, such as modal classification and effective refractive index retrieval in waveguides^{56,57}, dimension reduction to reveal underlying physics⁵⁸ and near-field manipulation of plasmonic nanoantennas⁵⁹.

As the plain form of the deep-learning model, MLP requires vectorized input and output.

an effective approach for many photonic tasks, the simple connections in the model still pose some difficulties when the intrinsic structure, target response and design space are either multimodal or hard to parameterize.

Such advanced models can extend the functionality and improve performance in many photonic design problems, which are discussed below.

In deep learning, a CNN is a class of deep networks that extracts the features of the inputs by using convolution operations on the output of each layer.

TCNNs are commonly implemented as a part of generative models for the discovery and design of photonic structures with large degrees of freedom.

For photonic design, RNNs are suitable to model optical signals or spectra in the time domain with a specific line shape originated from various modes of the resonance.

Network systems hybridizing CNNs and RNNs are promising techniques to model and design photonic devices that manipulate unconventional spatiotemporal properties of light.

Empowered by deep-learning algorithms, deep generative models can produce data that replicate in the same way as or are similar to the training dataset.

AEs are used to reduce the dimensionality of the design space and optical response features of photonic devices, providing essential information on the light matter interaction for device optimization⁵⁸.

Besides the standard architecture of the GAN and VAE, some other varieties of generative models can also be implemented for the discovery and design of photonic structures.

Other examples of generative models applied in photonics research have also been reported. For example, U-net, a deep neural network architecture originally designed for image segmentation, has been used to accurately predict the near-field optical responses of arbitrary 3D nanostructures, enabling the approximation of various far-field responses of nanostructures⁷⁵.

Recently, linear principal component analysis has been used to map and characterize a multiparameter design space of a photonic system⁸³.

Discussions and perspectives

In this section, we will comment on the directions, challenges and perspectives of interfacing deep learning with photonics.

deep learning can serve as a powerful tool in finding complex relations between a structure and its optical responses.

Spectra can be conveniently discretized into vectors⁴³ (single spectrum) or matrices⁴⁴ (multiple spectra), and thus can be readily incorporated into the deep-learning model.

To aid in the complete control of light, one immediate task that we need to work on is to enhance the capability of deep-learning models with more degrees of freedom, such as the phase⁷⁴, angular momentum, trajectory, nonlinearity, topology and near-field distributions⁷⁵.

Reinforcement learning is another approach worthy of further exploration, to expand the photonic design capability⁵⁰.

This indirect optimization process balances the current knowledge and unknown domains, and could potentially investigate more optical properties of photonic design tasks to

realize complete control of light.

The processed data from unsupervised learning strategies can be analysed to unveil the structural parameters of the photonic devices that have most impact, assisting the neural network to recognize the essential information of the design without redundant data collection^{90,91}.

Massive data collection can also be mitigated by merging the deep-learning models with underlying physics.

For example, instead of training a model with enormous datasets to approximate the bidirectional relationship between physical structures and their responses, deep learning can be adopted as intermediate steps to efficiently solve partial differential equations that describe the physical systems^{87,88,92}.

Furthermore, it is possible to generate a solution that satisfies certain partial differential equations by physics-informed deep-learning models⁹⁴.

From the viewpoint of efficiency, a well-trained deep-learning model is faster by orders of magnitude than traditional optimization algorithms⁴⁶, on the premise that enough training data are collected in advance.

The difference between data collection for deep learning and that for numerical optimization algorithms lies in how the models make use of data.

For deep learning, the model actively accumulates training data that holistically describe the problem under investigation, so that the model is driven by data in the sense that the quantity and quality of pre-collected training data contribute to its performance.

In contrast, traditional numerical algorithms passively produce data in the iterative optimization steps, where data, more like a by-product, are mainly used to check how far the optimization has proceeded and when the strategy needs to be changed. Usually, for a specific task, the quantity of data required by deep learning is much more than produced during a conventional optimization.

However, data collection for deep learning is a one-time cost, as opposed to an ongoing cost as in the case of ab initio numerical optimization.

This feature makes deep learning especially competitive in photonic design tasks of periodic structures such as metamaterials or photonic crystals, where a huge number of unit cells need to be optimized to constitute a device or a system.

Meanwhile, balancing training time and run time for a deep-learning model requires efficient exploitation of data, probably in an online manner where data are continuously provided as the model evolves according to different tasks.

To some extent, deep learning has similarities with gradient-based optimization methods, because the optimization of the deep-learning model is also directed by the average gradient of the loss function with respect to all the training data.

On the other hand, similar to gradient-free optimization that introduces more stochasticity to enlarge the possible design space, deep learning uses a large amount of pre-collected training data to extend the scope of design varieties far beyond what is achievable with optimization methods.

In conjunction with traditional numerical optimizations, a fast deep-learning approximator for the following optimization steps can be constructed by saving the simulation data right from the beginning of the optimization^{95,96}.

On the other hand, deep learning can help to identify a global minimum in various ways.

Latent variable models and generative models are able to capture the underlying physics so that unnecessary exploration of the solution space can be avoided. In more practical cases, traditional optimization and data-driven methods can be jointly incorporated into the same design pipeline, where deep learning serves to identify solutions near global minimum and traditional optimization refines the performance to the extreme.

Even if a true global optimum can hardly be found in a complex photonic design task, these optimization methods, including deep learning, always yield fairly good results beyond empirical designs.

However, because deep-learning and machine-learning models always require sufficient training datasets, data-driven approaches may not be an effective starting point in extreme instances where retrieving a single data point takes from hours to days.

Instead, direct optimization such as Bayesian optimization could be considered before deep-learning models are applied. Modern multidisciplinary fundamental science and engineering challenges, such as those related to energy, security, data processing and storage, as well as emerging quantum technology, requires multifunctional devices with highly constrained electromagnetic, chemical, thermal and mechanical properties.

Thus far, optical technologies have mainly used photonic optimization in a restricted design space that is largely limited to structural topology (geometry) and shape.

If temperature-dependent dielectric permittivity functions of the constituent material platform are incorporated into the training process, a globally optimized design with robust emissivity response over a wide range of operating temperatures may be determined. Global optimization search within the extended compressed space will not only ensure optimization of the design topology but will also provide optimal material properties and composition, or even guide the fabrication process to achieve the best possible performance of the targeted photonic device, architecture and system.

While the rapid advances in deep learning provide radically new approaches to solve photonic design problems, this assistance is not one-way but interactive⁹⁹.

Big data, the catalyst of deep learning, continues to revolutionize AI research with record-breaking improvements in various domains, while at the same time it causes an exponential increase in computational power and energy consumption.

At present, most deep-learning algorithms are deployed in conventional computers with von Neumann architecture, whose serial nature is an intrinsic barrier to efficient support for neural networks.

Even with some application-specific integrated circuits that are deliberately optimized to run deep-learning models, most industrial-grade models for practical applications still take a formidable time and cost to train. Superior to their electronic counterpart in both speed and power consumption, photonic platforms for deep learning, including nanophotonic scatterers¹⁰⁰, integrated silicon photonic chips^{101 103} and 3D-printed diffractive layers¹⁰⁴, are under active investigation.

The tremendous acceleration of deep-learning models on an optical platform depends on the capability of parallel signal processing of light, which makes matrix vector multiplications in constant time with respect to the matrix dimension, in contrast to the quadratic time complexity on a digital processor.

The interactions between new photonic structures and deep learning may overcome the limitation of current computing approaches and systems, and potentially lead AI research to new horizons.

The potential of new photonic structures, some of which may be designed by deep learning, to enable unconventional computing and AI techniques is worthy of further exploration.

summary

The concept of deep learning has gone far beyond being a computational analogy of biological neural systems and has emerged as a powerful tool that solves extremely complex problems by building up multilevel abstraction of massive data.

For the photonics community, deep learning and other AI techniques are currently transforming the areas of optical design, integration and measurements.

Deep-learning techniques have already demonstrated their tremendous potential for photonic structure design, optimization of architecture, materials and entire optical systems, and will continue to uncover new ways to unparalleled speed-up of optical measurements and even unlocking new optical effects.

In this Review, we have surveyed various model structures for photonic design, ranging from individual plasmonic nanoparticles to metamaterials comprising an array of meta-atoms, and to integrated photonic devices.

Such cross-disciplinary approaches merging AI, photonics and materials platforms will allow for large-scale photonic designs with unique functionalities as well as new methods for optical characterization, paving the way to high-speed super-resolution imaging, real-time detection and manipulation, efficient energy conversion systems and transformative advances in the area of quantum measurements and metrology.

On this path, the photonics community should ultimately build an optical structures and materials genome to construct a comprehensive dataset of photonic concepts, architectures, components and photonic materials to enable hierarchical machine-learning algorithms that could provide ultimate-efficiency devices.

This effort should be extended to realize all-optical platforms to perform deep learning and other AI algorithms at the speed of light, ushering in an even brighter AI era.